Multi-Modal Classical Machine Learning Framework for Integrated Heavy Rainfall Prediction and Hydrodynamic Flood Inundation Modeling

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Problem Statement:
Extreme rainfall events and sudden flash flooding present critical threats to urban infrastructure, economy, and human lives, yet conventional numerical weather prediction (NWP) models often lack the local spatial resolution required for real-time flood response. Classical hydrodynamic physics models can accurately map surface inundation, but their computational complexity renders them too slow for rapid early warning systems during live storms. Furthermore, raw meteorological data and terrain topographies operate on disparate spatial and temporal scales, making unified data fusion a major technical bottleneck. Existing simplified prediction systems frequently suffer from high false-alarm rates or fail to account for non-linear terrain physics like topographic wetness and drainage capacity. There is an urgent need for a computationally efficient, end-to-end machine learning pipeline that fuses multi-modal meteorological, satellite, and elevation data to deliver real-time precipitation forecasts and flood depth estimates without relying on heavy deep learning architectures.

Objective:

- Spatio-Temporal Precipitation Forecasting: Construct a multi-model ensemble (XGBoost, LightGBM, CatBoost) to perform 0–6 hour heavy rainfall prediction ($\geq 64.5 mm/day$) by processing engineered temporal lags and spatial neighborhood features.
- Topographic Inundation Modeling: Develop an ML surrogate model (Random Forest / Gradient Boosting Regressor) integrating predicted precipitation volumes with DEM-derived terrain metrics (Slope, Topographic Wetness Index) to estimate real-time surface flood depth.
- Explainability & Interactive Deployment: Implement TreeSHAP for feature attribution and build a production-ready Streamlit dashboard with interactive map layers displaying precipitation heatmaps, inundation extent, and risk alerts.

Dataset:

- Dataset name/source:
    - ERA5 Hourly Reanalysis Data (ECMWF via Copernicus CDS API)
    - NASA Global Precipitation Measurement (GPM IMERG) Satellite Rasters
    - USGS Shuttle Radar Topography Mission (SRTM) 30m Digital Elevation Model (DEM)

- Link:

https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels?tab=download
https://gpm.nasa.gov/data/imerg
https://www.usgs.gov/centers/eros/science/usgs-eros-archive-digital-elevation-shuttle-radar-topography-mission-srtm

Approx. number of samples:

$\sim 100{,}000 +$ spatiotemporal grid records across a multi-year historical timeframe

Key features:

- Meteorological Variables: Precipitable Water Content (kg/m^2), Relative Humidity , Atmospheric Pressure Tendency ($\Delta P/\Delta t$), u and v Wind Vector Components, Temperature ($2m$), Surface Runoff.
- Engineered Spatio-Temporal Lags: $1h$, $3h$, $6h$, and $24h$ Rolling Precipitation Sums, Spatial 3×3 and 5×5 Grid Kernel Reflectivity Means and Variances.
- Topographic Hydrology Features: Terrain Elevation (Z), Local Slope Angle (β), Flow Accumulation (α), Topographic Wetness Index ($TWI = ln\,(\alpha/tan\,\beta)$), Distance to Drainage River Networks (D).

Target variable:

- Primary Target: Binary Heavy Rainfall Indicator (1 if Rainfall $\geq 64.5mm/day$, else 0) for early warning hazard detection.
- Secondary Target: Surface Inundation Depth ($meters$) for predicting localized surface water accumulation.

ML Task & Proposed Approach:

- ML Task: Supervised Classification (Heavy Rainfall Alert) and Supervised Regression (Flood Inundation Depth Estimation).
- Proposed ML Algorithms/Models: XGBoost, LightGBM, Random Forest, and CatBoost.
- Brief Description of How ML Will Be Applied:
    1. Data Integration: Meteorological parameters from ERA5, precipitation estimates from NASA GPM, and topographic elevation features (Slope, Topographic Wetness Index) derived from SRTM DEM are merged into a single spatially aligned tabular feature matrix.
    2. Feature Engineering: Temporal rolling sums ($1h$, $3h$, $6h$) and spatial neighborhood aggregation features are generated to capture storm dynamics and terrain drainage capacity.
    3. Model Training & Resampling: Tree-based ensemble models (XGBoost, LightGBM, Random Forest) are trained to classify extreme rainfall events and predict flood depth. Class imbalance for rare heavy rain events is handled using SMOTE and cost-sensitive class weighting.
    4. Validation & Interpretability: Models are evaluated using Time-Series Block Split cross-validation, and SHAP (SHapley Additive exPlanations) is applied to provide feature attribution for predictions.

Evaluation Metrics:

- **Classification Metrics (Heavy Rain Hazard Detection):**
    - Critical Success Index (CSI / Threat Score)
    - Probability of Detection (POD / Recall)
    - False Alarm Ratio (FAR)
    - Precision, F1-Score, and Precision-Recall AUC (PR-AUC)
- **Regression Metrics (Inundation Depth Estimation):**
    - Root Mean Squared Error (RMSE)
    - Mean Absolute Error (MAE)
    - Coefficient of Determination (R^2 Score)

Expected Outcome:

The project will deliver an integrated classical machine learning pipeline that accurately predicts heavy rainfall events ($\geq 64.5mm/day$) and translates those forecasts into surface flood depth estimates. The system will achieve high predictive performance on severe weather events (evaluated via Critical Success Index) while bypassing heavy deep learning computation. The final submission will feature an interactive Streamlit

dashboard displaying interactive map visualizations and SHAP feature attribution plots to explain how topographic and atmospheric variables drive flood predictions.